Manual

MSA - Measurement System Analysis

PMV-MSA 8.2

Version 1.0.23049

Last changed on: 02.09.2020

Copyright

Contents

1 Measurement System Analysis – MSA

Purpose

You use the product *Measurement System Analysis - MSA* to perform a measurement system analysis according to the type 2 and type 3 gage study. This analysis is part of the test equipment management.

Implementation notes

You can use the product *Measurement system analysis - MSA* to identify statistical values for measuring equipment. Using these statistical values, you can assess the capability of measuring equipment for specific inspection processes.

Integration

The product *Measurement System Analysis - MSA* is part of the test equipment management. The inspection planning is the basis for the execution of a measurement system analysis. The measurement system analysis itself is performed as part of the calibration inspection requirement. You call the analysis in the toolbar of the inspection step characteristics of a calibration inspection requirement.

Features

The following functions are available:

- Extension of the inspection planning to perform measurement system analyses
- Identification of statistical values for the measurement system analysis to prove that the test equipment is suitable (variations according to measuring instrument or inspector, comparability, variations from part to part, etc.).
- Integration of type 2 and type 3 gage study

2 Inspection planning

Overview

Menu	Quality management → In-production inspection → Inspection planning Quality management → Goods receipt → Inspection planning Quality management → Goods issue → Inspection planning Quality management → Initial sample inspection → Inspection planning Quality management → Gage management → Inspection planning
Transaction code	iplp[1]
Function authorization	iplp

Available user fields		
Where	**Object type/user field key**	**Source (type)**
Inspection plan: Table and detail view	Production: CPPL/FEP Goods receipt: CPPL/WEP Goods issue: CPPL/WAP Initial sample: CPPL/EMU Gage management: CPPL/PMV	QM
Inspection plan characteristics: Table and detail view	Production: CPPLMM/FEP Goods receipt: CPPLMM/WEP Goods issue: CPPLMM/WAP Initial sample: CPPLMM/EMU Gage management: CPPLMM/PMV	QM
How to configure user fields?	Which user field types are available?	

Purpose

A careful inspection planning is usually the most important function, as inspection plans are the basis for the generation of inspection orders and the resulting inspections. This is true for inspections in the goods receipt process, during production and in the goods issue process. It also applies for the inspection of initial samples and the calibration of test and measuring equipment. As the requirements of an inspection planning are almost identical in all areas, the HYDRA inspection planning functions are almost identical for all these areas.

In general, master data catalogs help the user to generate inspection plans. Therefore, careful maintenance of master data is very important.

[1]If you start the application *Inspection Planning* using the transaction code, you cannot create a new inspection plan.

Integration

You require the inspection plan to generate inspection requirements, inspection steps and the resulting inspections/calibrations.

The generated inspection requirements include the used inspection plan. Also some evaluations use the inspection planning You can thus expand the basic data of a control chart of an inspection order characteristic by the data of the same inspection plan number. The same is possible for the global control chart evaluation. Here, you can use specific filter fields of the inspection plan. The global control chart evaluation does not integrate calibration data and thus you cannot filter by calibration inspection plans.

Another essential relation is established to the production control plan. You cannot create a production control plan without having generated inspection plans beforehand.

Requirements

To create inspection plans, you must maintain/edit the relevant master data, The identification of the master data, that you must maintain, depends on the respective application. However, it is a fundamental to edit the article and the characteristics beforehand. If you create inspection plans for article groups, you must first create the article groups.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt and goods issue. By default, the following areas are available:
In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

Active

Enable this option to narrow down the list of inspection plans to view only active inspection plans. If this option is disabled, the list only includes inspection plans with a status "in process" or "released". The third state of this option (grayed out) shows all inspection plans. This is the initial state.

Article number

Filters the article number in case of inspection plans based on articles.

Article designation

Filters the article designation/name in case of inspection plans based on articles.

Article group

Click the icon  to call the article group tree, if you want to filter the inspection plan for an article group. There is a function to accept and cancel the activity.

Operation

Operation number

Customer number

Enter the number directly or call the customer catalog and select an entry to take it over.

Supplier number

Enter the number directly or call the supplier catalog and select an entry to take it over.

Manufacturer number

Enter the number directly or call the customer catalog and select an entry to take it over.

Field descriptions

Inspection plan header

Area, inspection plan number, inspection plan index

The "Area", "Inspection plan number" and "Inspection plan index" uniquely identify the existing inspection plans. Select the area. Enter alphanumeric characters for the inspection plan number and inspection plan index. All three fields are mandatory fields.

The input in these three fields must be unique, i.e. no other inspection plan may exist that already includes this information. If you assign a structured inspection plan number, you can include specific information. This information might be helpful later on for sorting. As an alternative, you can also use the article number or the number of the test equipment group as inspection plan number. The inspection plan index corresponds to the inspection plan version. If you must modify an existing inspection plan version, but it is not possible to change the inspection plan as it has already been used for the generation of an inspection order, you must copy the original inspection plan and change the inspection plan index (e.g. increment by one). You should keep the inspection plan number with regard to subsequent evaluations, if possible (e.g. display of order-independent control charts where the inspection plan versions are different and the inspection plan number is identical).

If you generate calibration inspection plans, you can only enter 24 characters into the field *Inspection plan*. Activity calendar: Depending on the system configuration, you enter the inspection plan number in the field *Project number.* There, the field *Project number* is restricted to 24 characters.

Inspection plan type

There are two types of inspection plans: "Inspection plan of articles" and "Inspection plan of groups". If you select "Inspection plan of articles", you assign an article from the article master data to generate an inspection plan for this article. If you want to generate an inspection plan for a group of articles, select the type "Inspection plan of groups". You can then select an article group. The selected article group is displayed in a separate field within the tree structure.

Article

Enter the article number. If you know the article number, you can directly enter it. If not, you can open the article catalog and identify and transfer the requested article using the filter and sort criteria. Select an article to take over the drawing issue number, the article designation, the customer article number and the drawing number from the master data record. The respective data is then displayed in the corresponding fields.

If the field *Article number* or *Article group* is shown in the test equipment management, you must not make an entry here.

Drawing issue number

The drawing issue number can be entered directly, just as it is the case for the article number.

If you directly enter the article number and the drawing issue number, the master data record is identified with this information when you save the data. The fields article designation, customer article number and drawing number are then populated and displayed.

The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

If the field “Drawing issue number” is shown in test equipment management, you may not enter any values here.

Operation assignment

You can select between two assignment types: "One insp. plan for each OP" and "One insp. plan for all OPs".

Assign the operation type to define if the inspection plan you want to create includes the characteristics of different operations ("one inspection plan for all OPs") or if the inspection plan only includes the characteristics of one operation. If you select the setting "One inspection plan for each OP", the fields "Operation" and "Operation designation" are shown. In any other case, operation details are assigned in the area of inspection plan characteristics. If possible, you should select the option “One inspection plan for all OPs”. You can then print all inspection characteristics of an article, although it includes several operations.

In case there is no assignment to a production order, you should use the option "One inspection plan for all OPs". The advantage is that you must only enter the "fictitious" operation once in the inspection plan header, and not for each inspection plan characteristic. This inspection plan method requires a suitable system configuration that must be carried out by MPDV consulting.

Operation, operation designation

If you use operations, you must only complete one of the two fields "operation" or "operation designation". If you want to automatically generate an inspection requirement/an inspection order when you log on an operation/production order, you must only enter the operation number in the "Operation" field.

Note: These fields are used as search criteria when you generate inspection orders later on. That is: If you only enter the designation of the operation in the inspection plan, you must include this and only this information when you generate an inspection order later on.

Also if there is no assignment to a production order (i.e. goods receipt and test equipment management), you must enter a "fictitious" operation (e.g. 9999). This is required for the generation of an inspection step later on.

IO (inspection order) + Inspection station

If you enable the option "An IO for each inspection station", the assignment of inspection stations is used when it comes to inspection plan characteristics. In this case, the system generates a separate inspection step for each inspection station if the CAQ system option 1157 is configured accordingly. The separate inspection steps are also created with identical operations. If you want to use the specific AIP inspection modes for the goods receipt, the inspection in the laboratory/measuring room and for calibration, you must select the option "An IO for each inspection station". Only with this configuration, the system includes the information on the planned machine/machine group later on when the inspection step is generated. It is not necessary to assign an inspection station in this case.

Released/Active

Shows whether the inspection plan is "released" or additionally has the status "active". If the inspection plan is released or active, the respective options are checked. An inspection plan is only released and activated, i.e. its status is changed, if you use the respective toolbar functions. You must release the inspection plan before it can be activated.

When you generate inspection orders at a later point in time, the system only uses active inspection plans.

Released by / on

Shows the HYDRA user who has released the inspection plan. The release date is displayed additionally.

Cavity assignment

You require the authorization "ipl_cav" to display this field.

You can select "None" or "Sample" to assign cavities.

If this field is not available, you require a new program version of this application.

An inspection based on cavities is only possible, if the inspection is based on inspection points and characteristics.

With the function extension for the production inspection, there is the additional option of a piece-related collection based on cavities.

If the inspection is based on samples and/or pieces, you cannot assign cavities.

Valid from/till

The field entries are for information purposes only.

Type of inspection

Characteristic-related or piece-related inspection

If the inspection is based on characteristics and the sample size is for example 5, the characteristic is checked completely first. If the inspection is based on pieces, the characteristic is changed every time a measured value is collected, as all characteristics of one piece are inspected.

You can only use the piece-related inspection, if you collect data with reference to inspection points. If you want to perform a piece-related inspection in the goods receipt process, you must change the sample-related inspection into an inspection based on inspection points.

An inspection based on cavities is only possible in case of a characteristic-related inspection. If the inspection is piece-related, you cannot assign cavities.

Action

"Creation of inspection step" or "Create and release inspection step"

Only inspection steps that have been released can be checked.

Customer / Supplier / Manufacturer

If you select a customer/supplier/manufacturer, this inspection plan only applies for the selected company. Consequently, the customer/manufacturer/supplier is a key field of an active inspection plan. If one of these fields is populated, the inspection requirement must also include a company entry. Only then, you can use this inspection plan to generate the inspection requirement.

Dynamic modification type

Characteristic-related, batch-related or none

You require the authorization "iriscp.dynamic" to display these fields.

Batch-related: Here, the transitional definition, which is used for the dynamic modification, is specified in the inspection plan header. The initial inspection severity is also specified in the inspection plan header. You additionally require the reference to the dynamic modification norm in the inspection plan characteristics. However, the selection is restricted. You can only select dynamic modification norms that reference the same inspection severity definition as the assigned transitional definition.

Characteristic-related: The inspection plan characteristics include the references to the transitional definition, the initial inspection severity and the dynamic modification norm. The inspection severity definition of the assigned transitional definition and the assigned dynamic modifications norm must match.

None: The "dynamic modification" function is not enabled for this inspection plan.

Transitional definition

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been selected. It provides the possible inspection severities and controls switching between the inspection severities.

Initial inspection severity

You require the authorization "iriscp.dynamic" to display this field.

This field is only available if a dynamic modification relating to batches or characteristics has been selected. It defines the inspection severity that is used to inspect the first goods received according to the basics of the dynamic modification history.

Initial sample form

Use this option to select the form type for the initial sample inspection. Currently, only the form type "VDA volume 2, 4th edition" is supported. Depending on the form type, a respective list of categories to generate an initial inspection is shown in the inspection plan characteristics.

Gage method

You can edit the field "Process acc. to" in case of a calibration inspection plan. You can select between "Standard" and "QS9000". If you select "QS9000", the additionally displayed fields (Reference, Acceptance and Amendment) are mandatory fields.

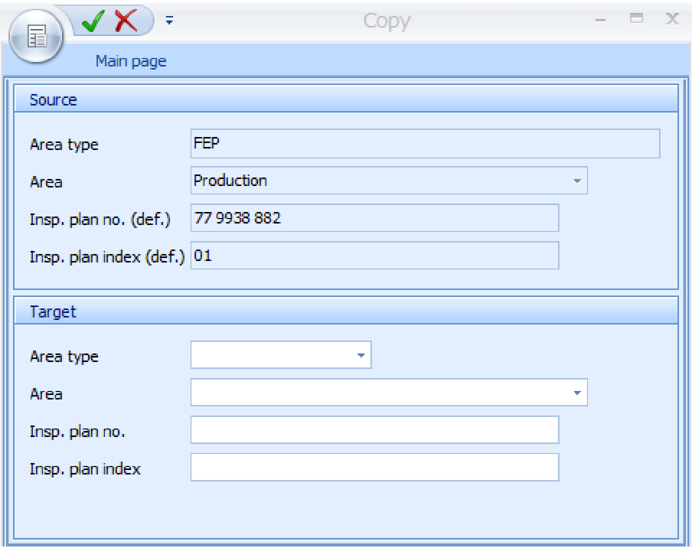
Editing functions

The key fields "area", "inspection plan" and "inspection plan index" cannot be changed in the editing mode.

Toolbar

Copy

To copy an inspection plan, the following dialog opens.



You can enter the target area type and the target area here. You usually select the values that are identical to the ones of the source inspection plan. Then enter the new inspection plan number and inspection plan index. If you generate a new version of an existing inspection plan, you normally use the identical inspection plan number and only change the inspection plan index.

If the option "Reorganization OP sequence" is activated, the characteristics are created in the new inspection plan with OP sequence numbers in increments of 10.

Activate

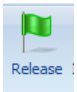
Function authorization: iplp.activate

Changes the inspection plan status to "Active".

Deactivate

Function authorization: iplp.deactivate

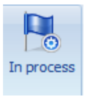
Changes the inspection plan status from "Active" back to "Released".



Release

Function authorization: iplp.release

Changes the inspection plan status from "In process" to "Released".



In process

Function authorization: iplp.inprocess

Changes the inspection plan status from "Released" to "In process".

Document management

Click here to call the Document management.

Import inspection plan

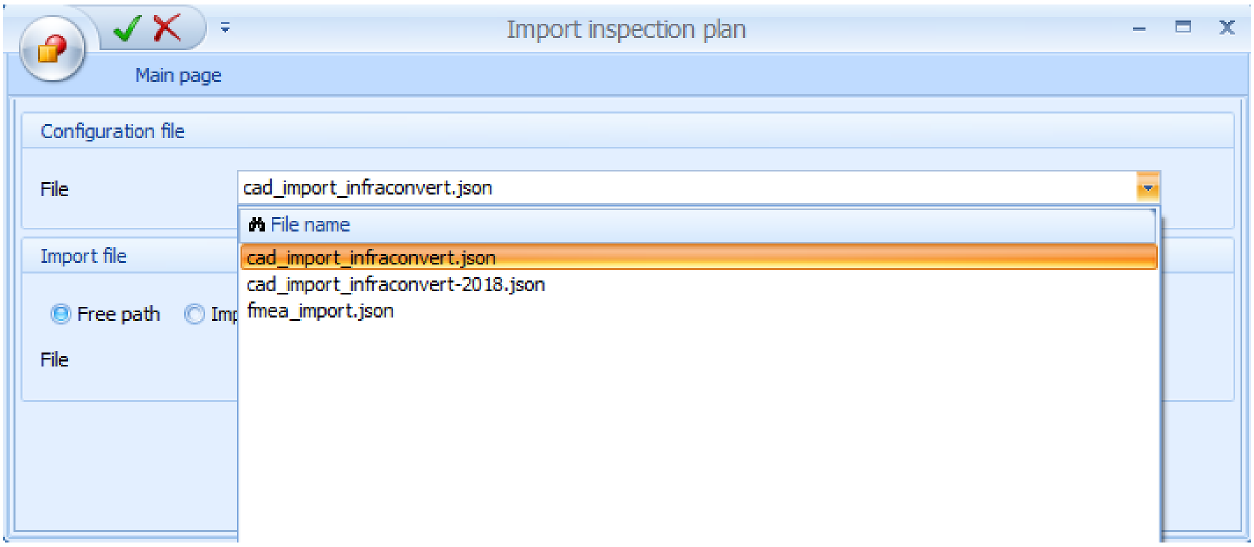
Function authorization: iplpimport

You can use this function to import inspection plan characteristics for the inspection plan selected. The system uses a previously created data file for the import.

"Import inspection plan" detail application

Function authorization	Iplpimport
License	FEP-PCF or WEP-PCF and EIS-CFM

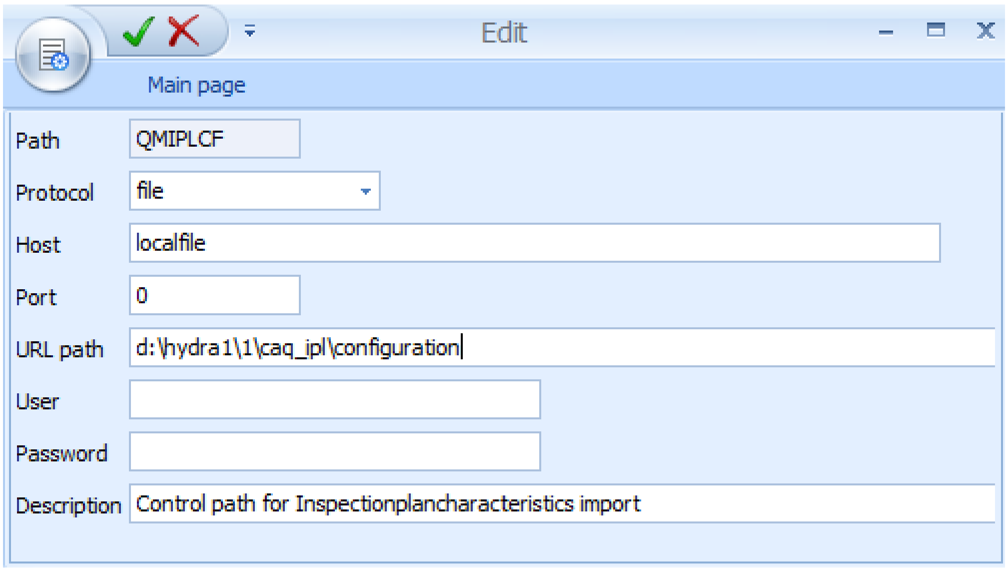
You can use the detail application "Import inspection plan" to add inspection plan characteristics to an inspection plan selected. You use this function if you have previously exported characteristic data into a data file using a CAD drawing or the HYDRA-FMEA application. In both cases, you first select the configuration file that you want to use for the import.



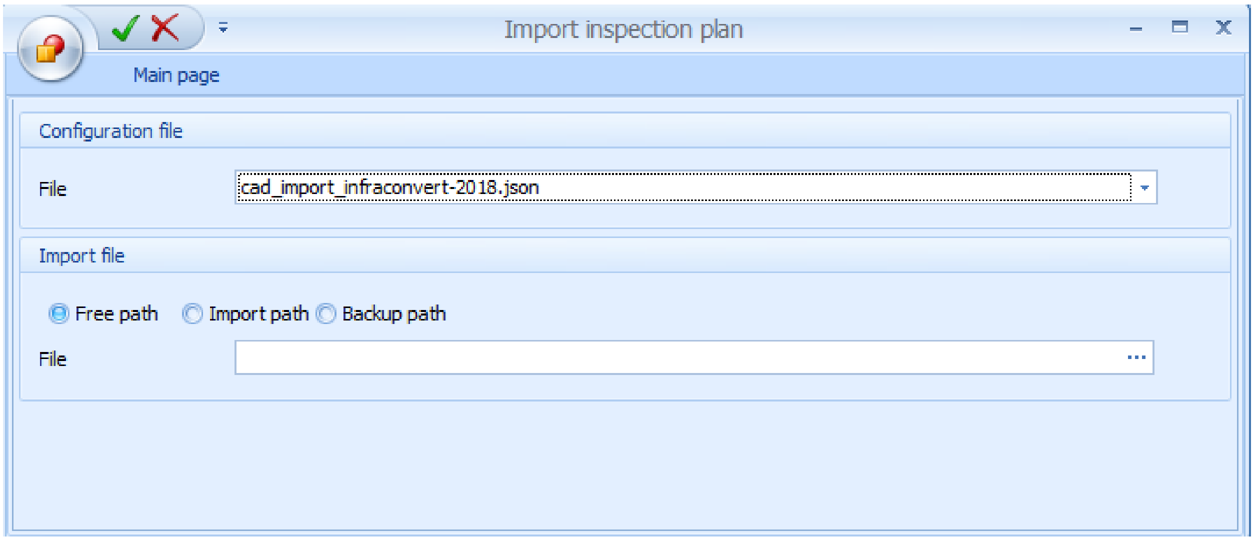
In the configuration file, you define the characteristic data that is transferred from the data file and you define the HYDRA characteristic fields where the data is transferred to. To import data from HYDRA-FMEA, you can use the configuration file "fmea_import.jason". To import data from a CAD drawing, the system provides the configuration files "cad_import_infraconvert.jason" and "cad_import_infraconvert-2018.jason" by default. These two configuration files are especially designed for the export of CAD characteristic data using the tool "infra CONVERT" or "infra CONVERT 2018" created by the company Elias GmbH.

If required, you can create custom configuration files. For details, refer to the procedure document "Configuration_InspectionPlanImport".

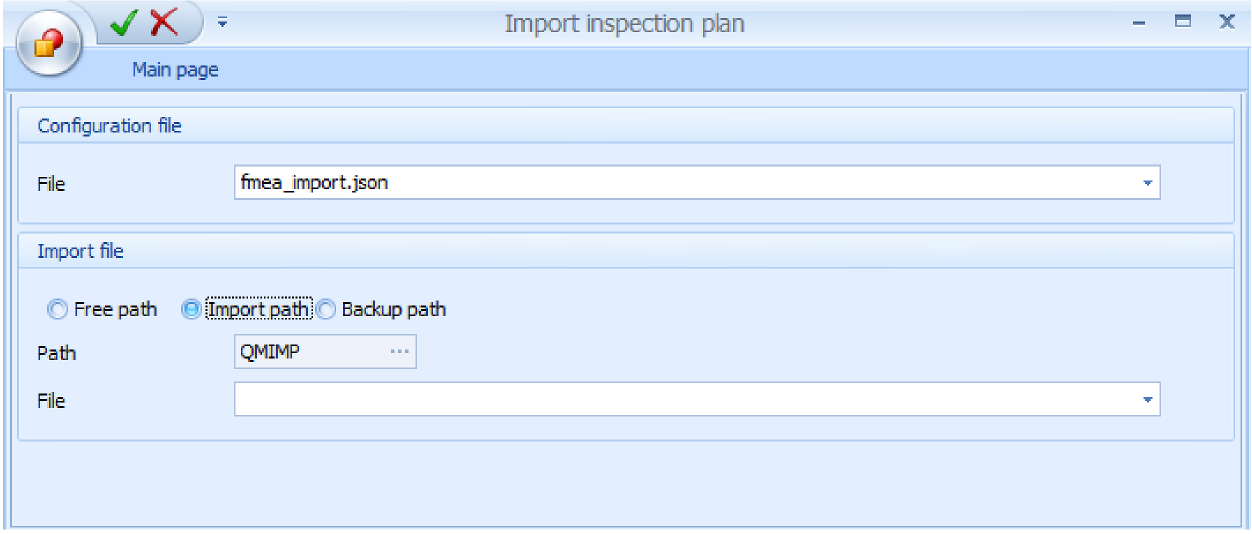
If the system does not display any configuration files, you must check the respective path configuration. In the HYDRA path configuration, the path "QMIPLCF" must be available. If the path "QMIPLCF" is not available, you must create the path. The below screenshot shows a possible path configuration for the configuration data.



When you have selected the configuration file, then you select the import file. For the import of data from a CAD drawing, enable the selection option "Free path". Using this selection option, you can select any import file.

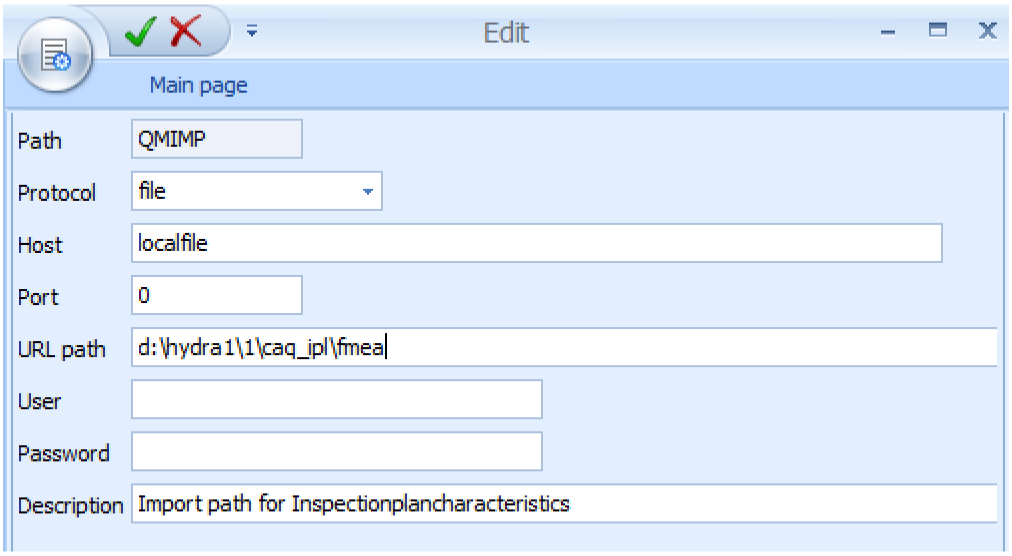


For the import of HYDRA-FMEA characteristic data, enable the selection option "Import path".



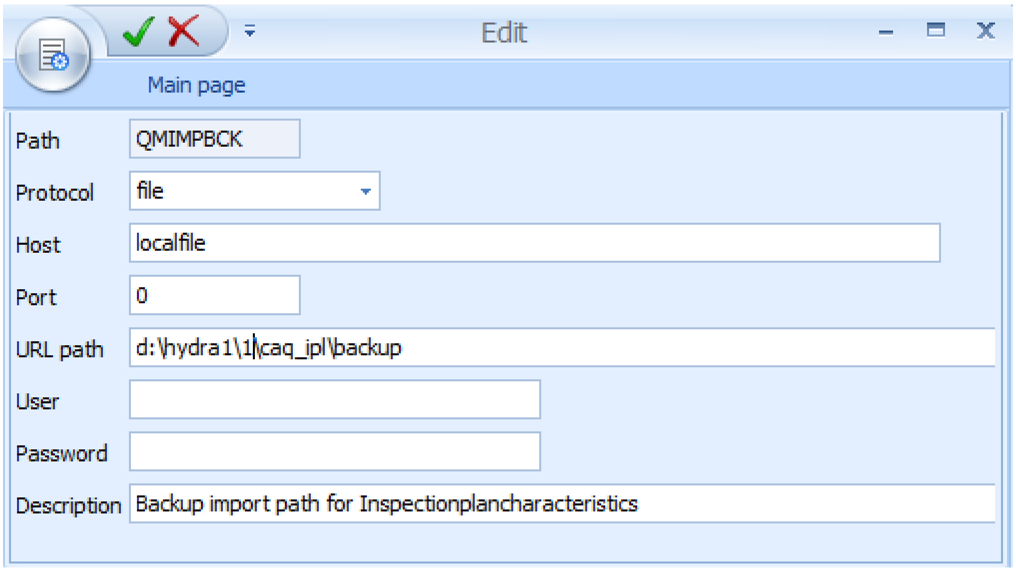
During the previous export of HYDRA-FMEA characteristic data, the exported characteristic data is automatically stored in the import path. In the HYDRA path configuration, the URL path is configured for the path "QMIMP". The path "QMIMP" must be available before using the import function for FMEA characteristic data. If the import path ist not available, you must manually create the path. If the import path is available, you must check the path configuration before the first import. If required, correct the specified URL path. It might be required to correct the system, for example.

The below screenshot shows a possible path configuration for the import of HYDRA-FMEA characteristic data.

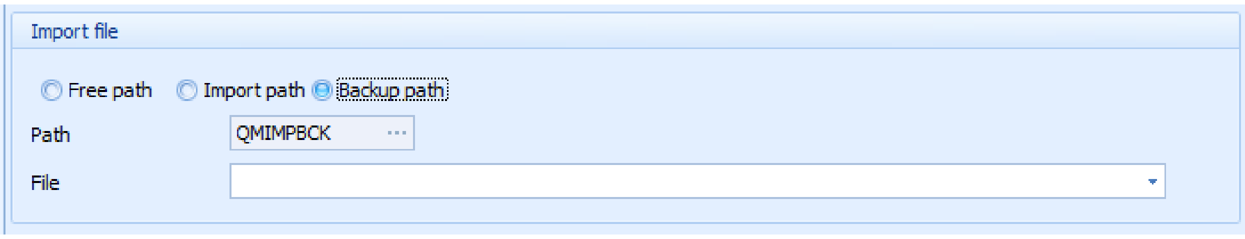


You must not only specify the HYDRA path, but you must also ensure that the specified URL path is available on the HYDRA server. If the specified path is not available, you must manually create the path on the HYDRA server. You must also check the specified URL path for all HYDRA path configurations mentioned below. If required, you must change the path configuration.

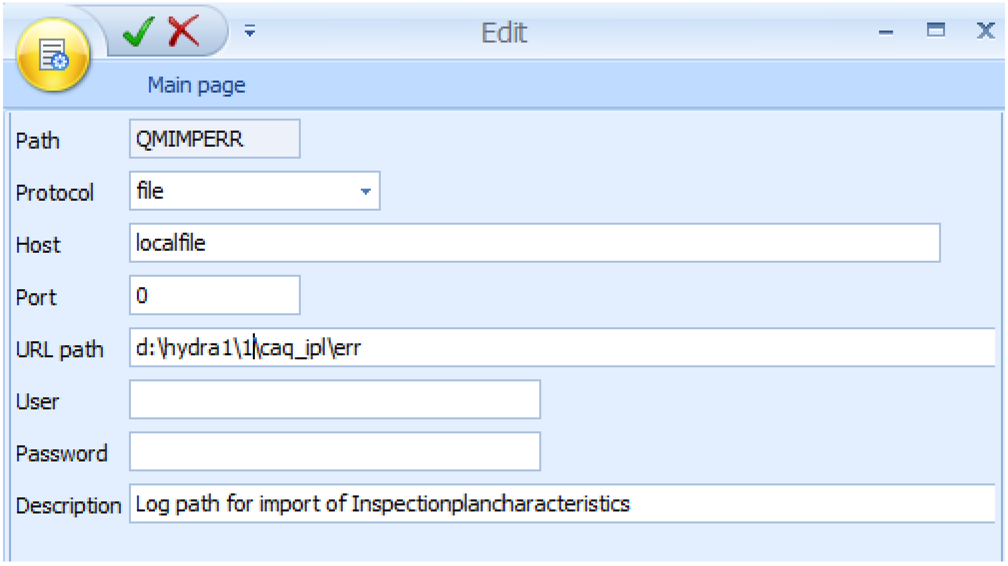
Before the first import, you must also create the backup path "QMIMPBCK" in the HYDRA path configuration because the data file is copied into the backup path after the successful import of HYDRA-FMEA characteristic data. The below screenshot shows a possible path configuration for the backup path.



If you want to re-import HYDRA-FMEA characteristic data after a successful import, you must enable the selection option "Backup path".



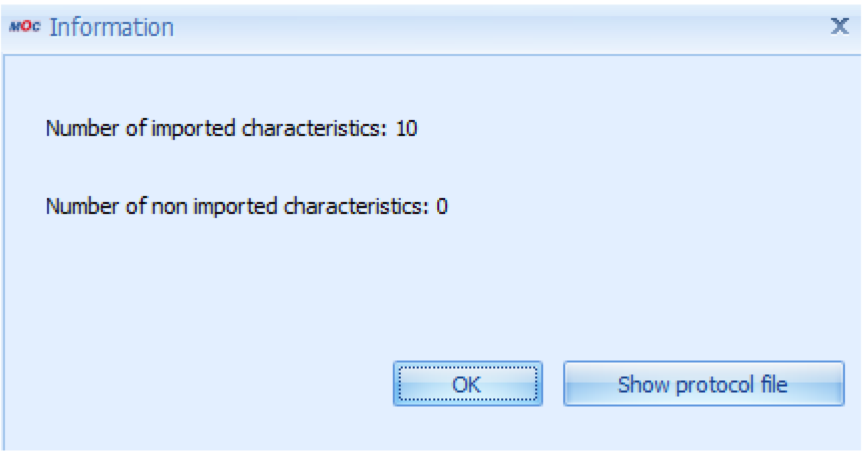
During the import process, a log file is created. To store this log file, you must create the path "QMIMPERR" in the HYDRA path configuration. The below screenshot shows a possible path configuration for the log path. If a valid log path is not available, you cannot import characteristic data.



For the import of HYDRA-FMEA characteristic data, you must finally select the file that you want to import.

The import of characteristic data is always completed using the button .

When the characteristic data have been imported, a message window shows the number of sucessfully imported characteristics and the number of not imported characteristics.



Use the button "Show log file" to open a log file. This log file specifies the characteristics that have been successfully imported and the characteristics that have not been imported.

"Print" – Detail application

Function authorization	iplp.print

The print dialog of the inspection plan header opens a list of available reports. These are Word forms. The web services that are available in the respective context determine the potential content of these forms. The form entries, i.e. the content of the list of forms of the respective print dialog, are defined within the master data of quality management. Here, the basics for new forms and the respective form properties are also specified. You require the respective license to be able to change the forms with respect to content and design.

Print – Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Inspection plan characteristics" – Detail application

The detail application *Inspection plan characteristics* is nearly identical to the application of the characteristics master data. Therefore, only additional features or modifications are described here.

Go to

For further information on the definition of characteristics, please refer to the functional description in the document MOC_CharacteristicsQM.

On the level of inspection plan characteristics, the inspection plan header defined beforehand assigns the respective inspection plan characteristics. To assign inspection plan characteristics, create a new data record, open the characteristics catalog and accept the selected characteristic. By accepting the characteristic, all master data entries are copied to the inspection plan characteristic. You can still change and/or complete each (copied) piece of information. The description of the characteristic is often completed to define it in more detail.

You can also create a characteristic that does not exist in the catalog of characteristics. But this is only recommended in exceptional cases, as all analyses (e.g. failure mode analysis) are based on characteristics included in the catalog of characteristics. Consequently, the characteristics catalog should be maintained.

You can define different properties and settings, before you complete the specific characteristic data.

Field descriptions

Inspection plan characteristics

OP sequence

The operation sequence number (OP sequence/AFO) specifies the inspection sequence. Entries must be unique. In the ideal case, operation sequence numbering should be assigned in steps of 10. You can then add a new characteristic between two existing characteristics at a later point in time.

Characteristic number

Number of the characteristic selected from master data.

Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. In case of an attributive inspection, the decision is often only based on the "pass" or "fail" results. Further characteristic types are the inspection chart and the information characteristic. The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

Mandatory inspection

If this option is activated, you must enter at least one measured value for this characteristic, before you can complete an inspection order including this characteristic.

Calculate characteristic

Identifies a characteristic to be calculated

Formula

Further details can be taken from the manual dealing with the CAQ master data characteristics.

Operation / Operation designation

this field are only available, if the setting "One inspection plan for all OPs" is activated in the inspection plan header.

If you want to include a reference to a productive operation in the characteristic, you need only store the operation number in the field "Operation".

If you define characteristics of QM operations, we recommend to add the operation. For characteristics of the same operation, this entry must always be identical.

If you define QM operations, you must be careful that the operation created in the production order later on is not the last operation. For example: If the last productive operation is the operation "0030" and you want to define characteristics with reference to a QM operation for this operation, you should use "0029" for this QM operation.

Initial sample creation

Selection list of categories for the creation of an initial sample inspection. The content depends on the form type selected in the inspection plan header. In the initial sample report, the inspection plan characteristics are grouped by the selected category for the creation and for each group a separate page is printed.

Copy characteristic

You can define here which characteristic is copied from an initial sample inspection plan into a production plan, for example. Only those characteristics are copied that have been specified as relevant in this field.

Details

This option specifies if - for this characteristic - the contents of the "details" tab are defined within the inspection plan or if they are identified using the available master data characteristic when an inspection requirement is later generated based on this inspection plan. If the "from characteristic catalog" option is selected, the "details" tab is hidden. The option "from characteristics catalog" is useful, if the same characteristic specifications must be defined for several articles (also for several article groups). In this case, specifications can be defined centrally within the master data. This reduces the input and update efforts considerably.

Specifications

This option specifies if - for this characteristic - the contents of the "specifications" tab are defined within the inspection plan or if they are identified using the available master data characteristic or an entry from the specification list when an inspection requirement is later generated based on this inspection plan. If you enable the option "from list" or "from characteristic catalog", the tabs "specifications", "chart 1", "default values chart 1", "chart 2" and "default values chart 2" are hidden. The option "from list" is only useful with an inspection plan for article groups. You use the specification list if the specifications for a specific characteristic are different for the individual articles.

No cavity

You require the authorization "ipl_cav" to display this field.

You can use this option to label a characteristic as not relevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

You must always enable this checkbox for attributive characteristics and inspection charts.

Dynamically modified

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. You can use this option to specify that a characteristic should not be modified dynamically, although the dynamic modification option is selected in the inspection plan header.

Transitional definition

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics has been activated in the inspection plan header. The selected transitional definition provides the possible inspection severities and controls switching between these different inspection severities. For the batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and in this case the transitional definition is not available on the level of inspection plan characteristics.

Initial inspection severity

You require the authorization "iriscp.dynamic" to display this field.
It defines the inspection severity that is used to inspect the first goods received according to the basics of the dynamic modification history. Only the inspection severities of the previously selected transitional definition are available. For the batch-related dynamic modification, the transitional definition is assigned in the inspection plan header and in this case the transitional definition is not available on the level of inspection plan characteristics.

Standard

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. The system uses the master data to assign the dynamic modification norm.

Inspection level

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. An inspection level can only be selected if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859.

AQL

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. You can only select an AQL value (Accepted Quality Limit) if the assigned norm corresponds to DIN ISO 3951 or DIN ISO 2859..

Method

You require the authorization "iriscp.dynamic" to display this field.
This field is only available if a dynamic modification relating to batches or characteristics is activated in the inspection plan header and if this characteristic is actually specified as relevant to dynamic modification. A method can only be selected if the assigned norm corresponds to DIN ISO 3951.

Sample group

You require the authorization "ipl_ipsampling" to display this field.

This field specifies which sample group this characteristic belongs to. The sample group specifies which characteristics are part of a sample/sampling characteristic.

Sampling

You require the authorization "ipl_ipsampling" to display this field.

If a characteristic has been assigned to a sample group, this option specifies if you use this characteristic to generate a sample. In this case, you call it a sampling characteristic. Activate this option to generate sampling characteristics. You must assign a sample group to the sampling characteristic.
If you want to inspect a sample using this characteristic, this option is not active.

If these fields are not available, you require a new program version of this application.

You must configure in the inspection plan characteristics, which are assigned to a sample group, that the details are provided by the inspection plan and not by the characteristics catalog, because you can only assign sample groups in the inspection plan.

All sample characteristics of one sample group must be included in the same inspection step. In the inspection plan, the combination of inspection station (machine, machine group) and operation must be identical for the sample characteristics.

If you want to combine the inspections for the producing machine (including sampling characteristic) and the inspections of the samples in one operation, you must consider the following issues when you plan the characteristics for this operation:

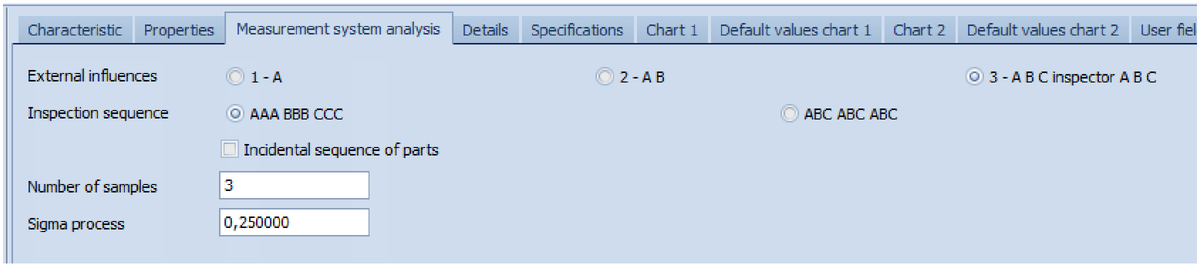
- The characteristics, which are used for the inspections of the producing machine, and the sampling characteristic must include a specific inspection station (e.g. MACHINE).
- The characteristics, which are used for the inspections of the samples, must be

assigned to a different inspection station (e.g. LAB).

- In the respective inspection plan, you must set the configuration "IO + inspection station" to "An IO for each inspection station".
- The terminal the producing machine has been assigned to must be configured specifically. For the above-mentioned example, you must assign the inspection station "MACHINE" in tab "QM functions".
- In the configuration of the terminal where the samples are inspected, you must assign the inspection station "LAB" in tab "QM functions" for the above-mentioned example.

Measurement system analysis

If the inspection plan is a QS9000 inspection plan, an additional tab *Measurement system analysis* is displayed.



Besides the sample size that you edit in tab *Specifications*, you must edit the key data of the measurement system analysis here.

Go to

The other fields of the respective tabs correspond to those of the characteristic master data and are outlined in the documentation entitled MOC_CharacteristicsQM.

Inspection plan characteristics – Toolbar

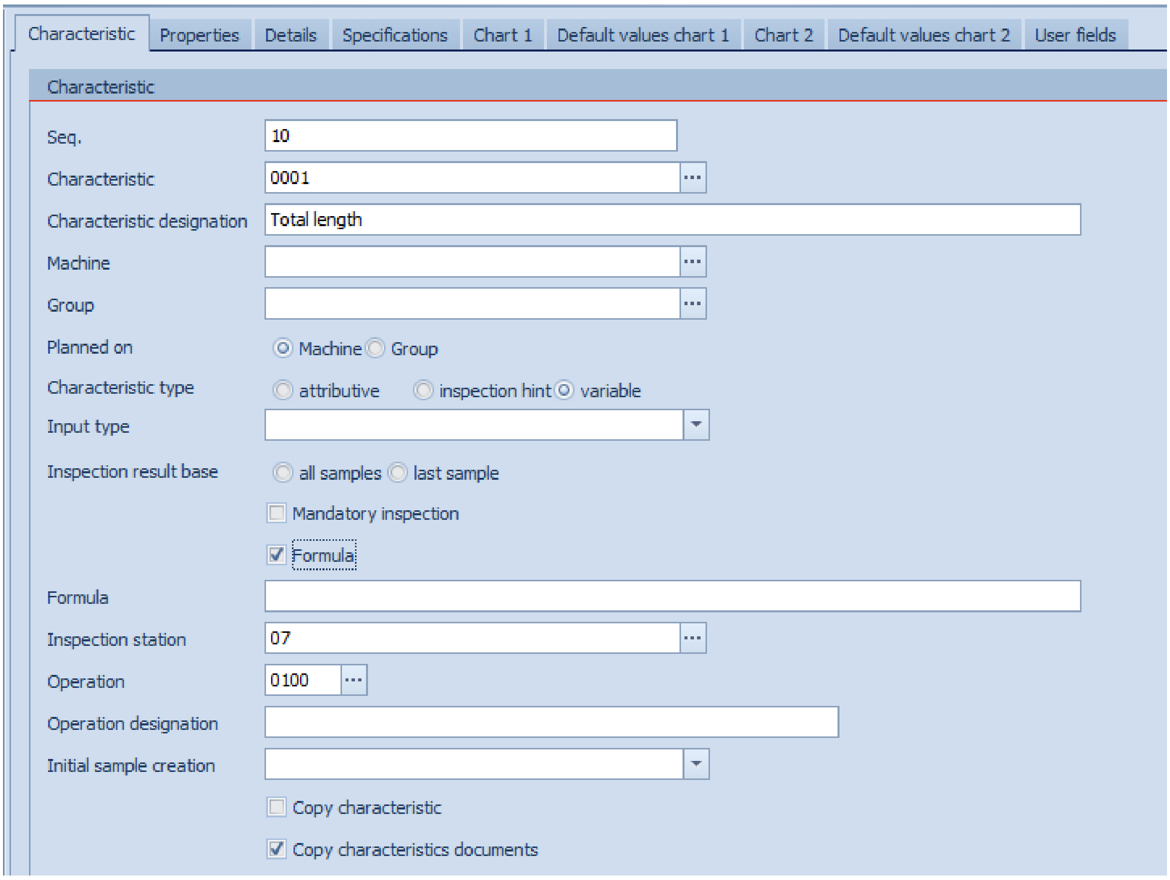
Create new data record

Select a characteristic from the master data. All characteristic details of the master data characteristic are transferred to the inspection plan characteristic.

When you select a characteristic from the master data, also the documents assigned to the master data characteristic are included in the inspection plan characteristic.

Copy

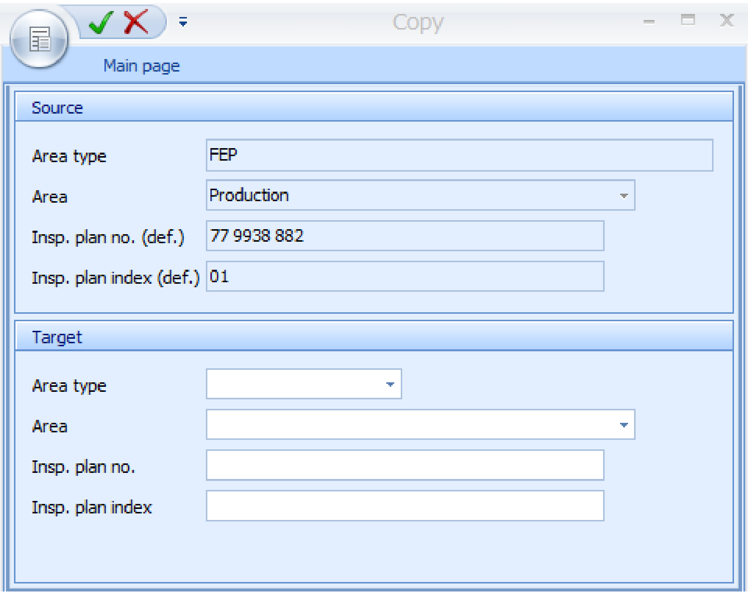
To copy inspection plan characteristics, the data of the selected characteristic are opened in insert mode. All fields can be modified. To save the data, you must assign an OP sequence number that does not yet exist in this inspection plan.



The documents of the inspection plan characteristic, for which the “copy” button has been clicked, are transferred.

Copy characteristics

Select a target inspection plan. The selected characteristics are then copied into the specified inspection plan. If the target inspection plan already includes characteristics, the new characteristics are created with OP sequence numbers in steps of 10.

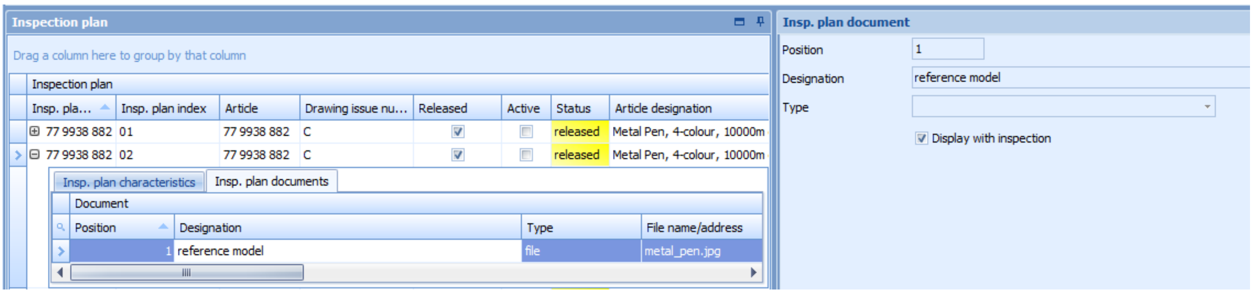


The documents of the different characteristics are also transferred.

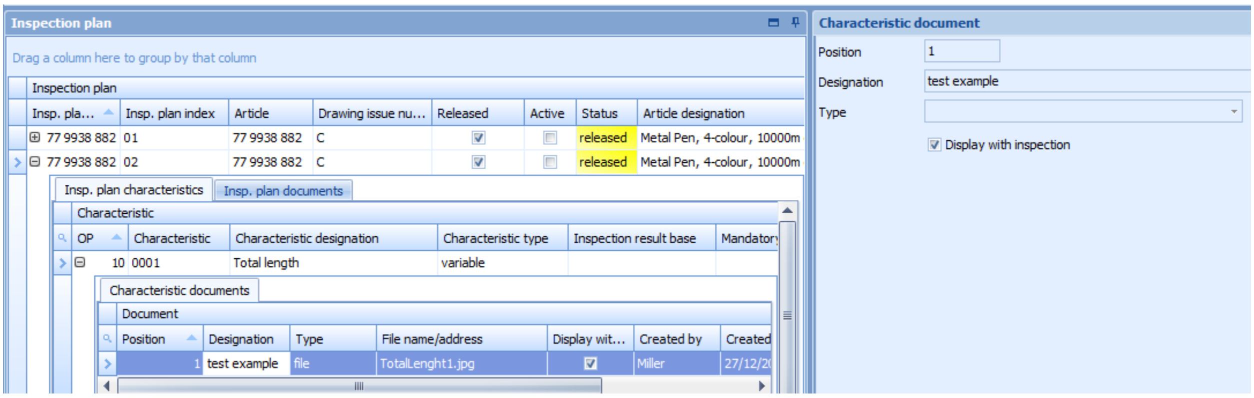
Document management

Click here to call the [Document management](#).

"Inspection plan documents“ and "Documents of inspection plan characteristics“ – Detail applications

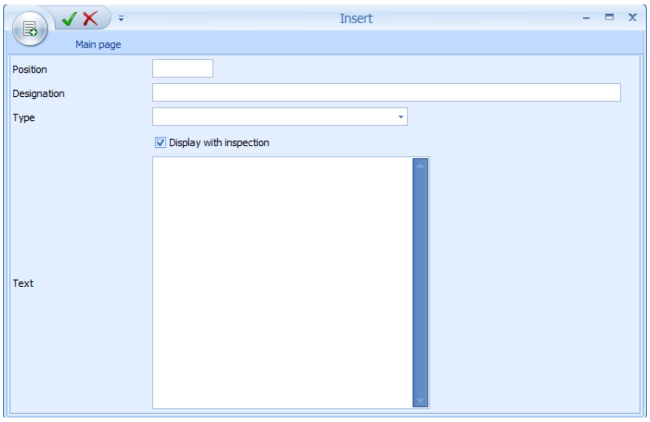


The above screenshot shows how an inspection plan document is assigned.



The above screenshot shows how a document of an inspection plan characteristic is assigned.

You can assign any number of documents to each inspection plan and each inspection plan characteristic, if the tab "Inspection plan documents" or the tab "Characteristic documents" has been activated in the master detail grid. If you activate these tabs, the toolbar provides the corresponding buttons to edit documents.



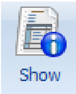
You can use all formats registered by Windows for the documents you want to assign. You can therefore assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The program link stored in Windows opens the documents.

The file types "File", "URL", and "Text" are available. The file name including the path may be entered manually for the "file" type. The file type “URL” allows to access the internet or intranet. Use the file type "text" to directly enter text.

You can assign a designation/name to each document that is added. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be distinct in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

"Inspection plan documents“ and "Documents for inspection plan characteristics" – Toolbar

In addition to the standard functions, the button to show documents is available.



Show documents

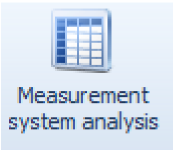
If a document link is stored, this button opens and shows the linked document. However, a program, which can show the linked file type, must be installed in the PC.

3 MSA (Measure System Analysis) - statistical values

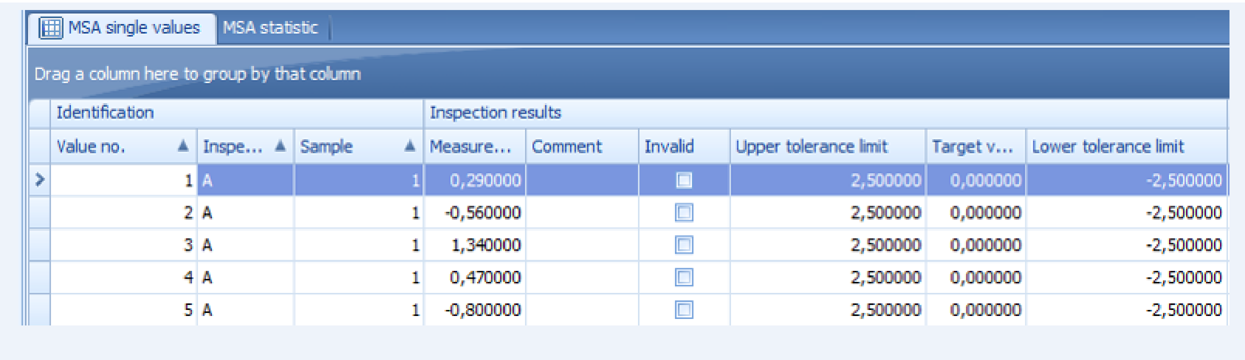
Overview

Menu	Not applicable, since the call is made exclusively from another application.
Transaction code	Not applicable, since the call is made exclusively from another application.
Function authorization	valuemsa

You can only call this application if you use the toolbar of the inspection step characteristics in a calibration inspection request. Use the following button in the toolbar of the inspection step characteristics:



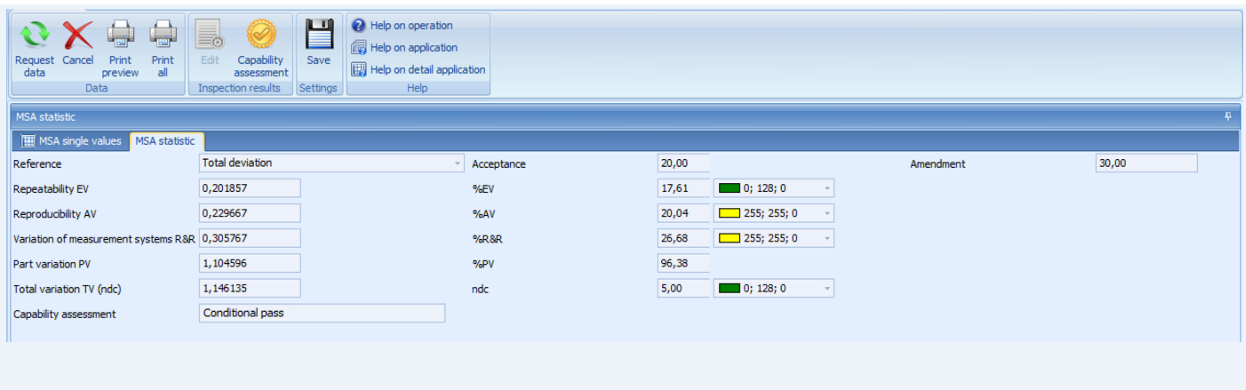
You can manage all inspection results for the measurement system analysis in the application MSA statistical values. Only the fields "Measured value" and "Comment" can be edited.



Identification			Inspection results					
Value no.	Inspe...	Sample	Measure...	Comment	Invalid	Upper tolerance limit	Target v...	Lower tolerance limit
1	A	1	0,290000			2,500000	0,000000	-2,500000
2	A	1	-0,560000			2,500000	0,000000	-2,500000
3	A	1	1,340000			2,500000	0,000000	-2,500000
4	A	1	0,470000			2,500000	0,000000	-2,500000
5	A	1	-0,800000			2,500000	0,000000	-2,500000

The user executes the input and assignment of inspector/sampling.

You can save a capability rating for the characteristic in the tab "Statistics MSA“.



Purpose

The application can execute a capability rating of test equipment.

A calculation of the MSA statistical value is done after all measured values are validated. This means all corresponding samples have been completed. The system does not carry out a statistic if not all measured values have been calculated.

Fields referring to these values will not be calculated if the calculation of individual components of the MSA is mathematically impossible.

The implementation is currently restricted to method 2 and 3 of the measured system analysis and can only be done for variable characteristics. The respective measured values shall be given in relative values.

Integration

Requirements

Field descriptions

Tab "Statistic MSA"

Reference

Taken from a superordinate inspection requirement

Acceptance

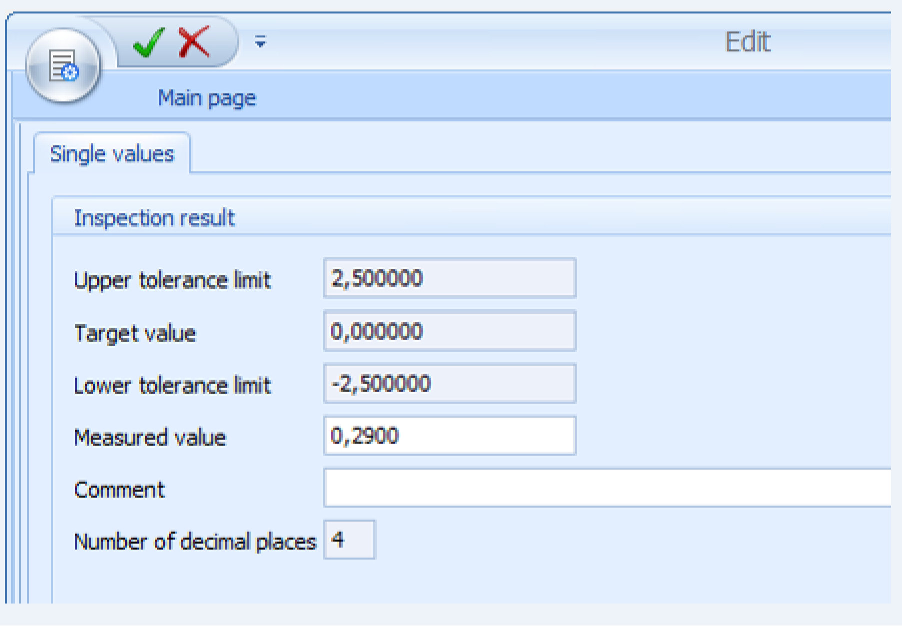
Taken from a superordinate inspection requirement

Amendment

Taken from a superordinate inspection requirement

Editing functions

The following dialog opens to edit a data record:



Toolbar

Capability rating

The capability rating takes the current measured system analysis values into the characteristic.